

More Complicated Logarithmic Equations (Part II) Homework

Solve.

$$\textcircled{1} \log_2 x + \log_2 (x-4) = 5$$

$$\boxed{x=8}$$

$$\textcircled{2} 1 = \log_4 (y-3) + \log_4 y$$

$$\boxed{y=4}$$

$$\textcircled{3} \log_3 x + \log_3 (x+8) = 2$$

$$\boxed{x=1}$$

$$\textcircled{4} \log_6 (x-8) + \log_6 (x-3) = 1$$

$$\boxed{x=9}$$

$$\textcircled{5} 1 = \log_5 (y+2) + \log_5 (y+6)$$

$$\boxed{y=-1}$$

$$\textcircled{6} \log_6 (b+4) + \log_6 (b-5) = 2$$

$$\boxed{b=8}$$

$$\textcircled{7} 3 = \log_2 (x^2 + 2x - 27)$$

$$\boxed{x=5 \text{ or } -7}$$

$$\textcircled{8} \log_6 (b^2 + 11b + 54) = 2$$

$$\boxed{b=-2 \text{ or } -9}$$

$$\textcircled{9} 0 = \log_4 (3x^2 + 11x - 3)$$

$$\boxed{x=\frac{1}{3} \text{ or } -4}$$

$$\textcircled{10} \log_2 (7y) + \log_2 (y - \frac{24}{7}) = 4$$

$$\boxed{y=4}$$

$$\textcircled{11} 1 = \log_3 (4x^2 - 5x + 4)$$

$$\boxed{x=\frac{1}{4} \text{ or } 1}$$

$$\textcircled{12} \log_3 (2y+3) + \log_3 y = 2$$

$$\boxed{y=\frac{3}{2}}$$

$$\textcircled{13} \log_5 b + \log_5 (b+3) = 1$$

$$\boxed{b=\frac{-3+\sqrt{29}}{2}}$$

$$\textcircled{14} 0 = \log_4 (5x) + \log_4 (x-2)$$

$$\boxed{x=\frac{5+\sqrt{30}}{5}}$$

$$\textcircled{15} \log_2 (x+1) + \log_2 x = 3$$

$$\boxed{x=\frac{-1+\sqrt{33}}{2}}$$

$$\textcircled{16} \log_5 y + \log_5 (3y+4) = 1$$

$$\boxed{y=\frac{-2+\sqrt{19}}{3}}$$

$$\textcircled{17} 2 = \log_{(x+2)} (7x+2)$$

$$\boxed{x=1 \text{ or } 2}$$

$$\textcircled{18} \log_{(x-1)} (6x^2 - 16x - 15) = 1$$

$$\boxed{x=\frac{7}{2}}$$

$$\textcircled{19} 2 = \log_{(x+3)} (2x^2 - x + 21)$$

$$\boxed{x=4 \text{ or } 3}$$